

Aditi

B.Tech Computer Science & Engineering (Artificial Intelligence & Machine Learning)

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Professional Summary

AI/ML Engineer and Computer Science undergraduate with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Data Analytics, and Software Development. Smart India Hackathon 2025 Grand Finalist with expertise in Python, TensorFlow, PyTorch, Scikit-learn, SQL, Linux, REST APIs, predictive modeling, feature engineering, and end-to-end ML pipeline development.

Education

Buddha Institute of Technology, Gorakhpur (AKTU)

2023 – 2027

Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning)

Relevant Coursework: Data Structures & Algorithms (DSA), Machine Learning, Deep Learning, Artificial Intelligence, Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, Software Engineering, Data Mining, Statistical Computing.

Technical Skills

Languages: Python, Java, SQL, Bash Scripting, C, C++, HTML, CSS

AI/ML & Advanced Paradigms: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Generative AI, Agentic AI, Federated Learning, Distributed Learning, Genetic AI (Evolutionary Algorithms), Large Language Models (LLMs), Prompt Engineering, Predictive Modeling, Feature Engineering, Sentiment Analysis, Computer Vision

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Gradio, TextBlob, VADER, Keras, Matplotlib, Seaborn

Tools & Infrastructure: Git, GitHub, Linux/Unix, PostgreSQL, MySQL, REST APIs, Cloud API Integration, Robotic Process Automation (RPA), Arduino Robotics, VS Code, Jupyter Notebook

Core CS Concepts & MLOps: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Software Engineering, Model Deployment, MLOps Fundamentals, Data Pipelines

Experience

AI/ML and Data Science Intern | Technex, IIT (BHU), Varanasi

Oct 2025 – Nov 2025

- Developed predictive and classification models using TensorFlow and Scikit-learn on real-world datasets, improving model performance and convergence efficiency.
- Automated data preprocessing, feature engineering, and Exploratory Data Analysis (EDA) workflows, reducing manual effort by 30%.
- Built **TypeFix**, an NLP-powered grammar and spelling correction system with automated error detection and intelligent content recommendations.
- Created statistical validation modules and visualization dashboards to improve model interpretability and analytical insights.

Summer Research Intern | IIT (BHU), Varanasi

May 2026 – July 2026

- Engineered end-to-end ML pipelines and deep neural network architectures using Python, PyTorch, TensorFlow, and Scikit-learn.
- Replicated **Swarmfed++**, replacing centralized parameter servers with Peer-to-Peer (P2P) particle swarm optimization (PSO) across edge nodes.
- Utilized high-dimensional Exploratory Data Analysis (EDA) and automated preprocessing to isolate signals from complex datasets, cutting bandwidth by 42%.
- Implemented mathematical optimization and statistical validation modules to ensure model transparency and training convergence.

Outreach & PR Head | GeeksforGeeks Campus Body, BIT Gorakhpur

Jul 2025 – Sept 2025

- Led outreach campaigns and institutional collaborations, increasing engagement and participation across technical initiatives.
- Organized coding workshops, hackathons, and industry interaction sessions benefiting 500+ students.

Projects

Audio Language Model (SIH 2025 Grand Finalist)

Python, PyTorch, NLP

- Developed an AI-powered voice interaction system using Natural Language Processing for intelligent query understanding and response generation.
- Recognized as a National Grand Finalist in Smart India Hackathon 2025 for innovation and real-world impact.

AI Social Media Caption Generator

Generative AI, Gradio, Python

- Built an LLM-powered caption generation platform achieving approximately 85% content relevance accuracy.
- Developed a scalable Gradio interface for real-time SEO-friendly content creation.

Mental Health & Mood Tracker

Sentiment Analysis, PostgreSQL

- Built an AI-powered mood tracking platform using VADER and TextBlob sentiment analysis techniques.
- Implemented secure PostgreSQL-based storage and trend visualization for personalized wellness insights.

Linux GPA Calculator

Bash Scripting, Linux

- Developed a command-line GPA/CGPA calculator using Linux shell scripting and dynamic credit-based computation logic.

Achievements & Certifications

- Grand Finalist – Smart India Hackathon (SIH) 2025 | Finalist – Hacknovate 7.0 (ABESIT Engineering College).
- Winner – TechSprint GNIOT GdG Hackathon | Winner & Runner-Up – TECH YUVA 9.0 (2024–25)
- Shortlisted for Innovation, Design & Entrepreneurship (IDE) Bootcamp, JIET Jodhpur (MoE Innovation Cell)
- Certifications: RPA Bootcamp (NIELIT Gorakhpur), AI Workshop (IIT Delhi TRYST), Machine Learning & Data Science Training
- Professional Memberships: Active Member of the **IEEE Student Branch (Uttar Pradesh Section)**